

Executive Summary



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Introduction and Context

In recent years, the increasing number of vocational (SMK) and university students undertaking internships at PT. Madani Intelsysdata (MID) has presented a practical challenge: how to manage and deliver structured learning materials efficiently for multiple interns with diverse backgrounds.

PT. Madani Intelsysdata, established in 2004, is a consulting firm specializing in Business Intelligence, Data Warehousing, and Reporting Systems. With a consistent record of hosting interns as part of its corporate learning culture, MID recognizes the need for a centralized digital platform that supports continuous learning, assessment, and communication between supervisors and interns.

The internship project undertaken by Mohamad Rasyedo Monoarfa, a student of Informatics, Faculty of Engineering and Informatics, Universitas Multimedia Nusantara, responds directly to this need. Acting as a Fullstack Java Developer, the intern designed, conceptualized, and implemented a web-based E-Learning Portal (MID E-Learning) that supports the learning process for incoming interns.

This executive summary outlines the background problem, project findings, and recommendations arising from the internship, focusing on how the solution contributes to MID's operational efficiency and long-term digital transformation goals.

Policy Problem: The Need for Structured Digital Learning

As part of its mission to provide high-value business solutions, PT. Madani Intelsysdata continually engages interns in various technical learning activities. However, before this project began, the learning process for interns was mostly manual and decentralized — relying on shared files, direct mentoring, and fragmented documentation.

This approach led to several inefficiencies:

- Lack of a standardized repository of training materials.
- Difficulty in monitoring individual intern progress.
- Limited interaction and feedback mechanisms.
- Repetition of onboarding and mentoring efforts for each new batch.

Given these issues, MID required a structured and accessible learning management platform that would streamline intern onboarding, learning, and evaluation while maintaining company standards of professionalism and technical quality.

Scope and Objectives

The internship project was designed to address this specific institutional challenge by developing the MID E-Learning Portal.

The objectives were to:

1. Translate the company's informal mentoring process into a structured, digital learning experience.
2. Provide an integrated platform where interns could access materials, follow courses, and track progress.
3. Reduce manual supervision time for system analysts and mentors.
4. Enhance overall consistency and efficiency in the company's internship program.

To achieve this, the project involved end-to-end development, including:

- Requirement analysis under the supervision of System Analyst Martio Chandra.
- UI/UX design using Figma and conceptual modeling using PlantText.
- Backend development using Java Spring Boot.
- Frontend implementation using Tailwind CSS.

The intern worked independently throughout the process, demonstrating initiative and adaptability in solving technical and conceptual challenges.

Key Findings and Learning Outcomes

The project revealed several key findings regarding both the company's needs and the development process itself:

- System Centralization Increases Efficiency

A centralized e-learning system reduces repetitive training sessions and provides long-term documentation for future interns. Supervisors can now reuse course modules, monitor progress digitally, and focus more on quality assurance.

- Independent Fullstack Development is Feasible for Internal Tools

Despite being executed by a single developer, the project showed that small-scale, company-specific tools can be effectively developed using modern frameworks like Spring Boot and Tailwind, minimizing the need for large external teams.

- Skill Integration Between Theory and Practice

The project integrated academic knowledge — such as Human Computer Interaction (for design principles), Web Programming (for frontend logic), and Object-Oriented Programming (for backend architecture) — into practical industry application.

- Challenges of Solo Development

Working without a team required strong self-management and problem-solving ability. The intern developed strategies for independent research, time discipline, and focus, which are key competencies in professional environments.

Overall, the project resulted not only in a functional learning platform but also in a deeper understanding of real-world software development cycles — from concept design to deployment.

Recommendations

Based on the project findings, several actionable recommendations are proposed for PT. Madani Intelsysdata to maximize the impact of the E-Learning system and enhance its internship program:

1. Adopt and Integrate the E-Learning Platform Company-Wide

Implement the portal officially as the standard onboarding and training system for all interns. This ensures consistency, scalability, and a structured learning experience across departments.

2. Expand the Platform with Administrative and Analytics Modules

Future development should include admin dashboards for monitoring user data, course completion rates, and learning analytics. This will allow MID to evaluate the effectiveness of its internship programs using measurable indicators.

3. Involve System Analysts and Mentors in Content Development

Encourage collaboration among senior staff to continuously update and enrich course materials within the platform, ensuring that the content reflects current business intelligence trends and technologies.

4. Continuous Improvement through Feedback Loops

Establish a periodic review mechanism where each batch of interns provides feedback on the system's usability and learning experience. This feedback can be used for iterative improvements and system enhancements.

5. Explore Potential for Broader Use

The system could be adapted for use in external training, certification programs, or partnerships with educational institutions, enhancing MID's reputation as a learning-oriented technology firm.

Conclusion and Call to Action

The development of the MID E-Learning Portal represents a strategic step toward digitizing and streamlining the company's internship learning process. By addressing inefficiencies in the previous manual system, this project not only supports internal knowledge management but also strengthens MID's commitment to innovation and continuous learning.

For PT. Madani Intelsysdata, adopting and expanding this system will:

- Improve operational efficiency in managing interns.
- Ensure knowledge continuity across multiple internship cycles.
- Reduce onboarding time and supervision workload.
- Reinforce the company's image as a forward-thinking, digitally capable organization.

For the intern, this project provided a transformative experience — applying theoretical academic knowledge to solve a real-world business challenge, demonstrating independence, technical mastery, and professional growth.

Moving forward, the company and academic institution can collaborate further to refine the system, enhance its scalability, and integrate it as a model for effective internship digitalization in the technology consulting sector.
